6. Product Measures.

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6.1 Multi-dimensional Lebesgue measure

An *interval* (or rectangle for $n = 2$, cube for $n = 3$) in $\mathbb{R}^n$ is a k-cell $\prod_{i=1}^n I_i$, where each I_i is a 1D interval.

6.2 Product σ-fields

Definition 6.1 (Product σ-field). The product σ-field of $\mathcal{F}_1$ and $\mathcal{F}_2$ is generated by the family of measurable rectangles:

$$\mathcal{F}_1 \times \mathcal{F}_2 \triangleq \sigma(\{A_1 \times A_2 : A_1 \in \mathcal{F}_1, A_2 \in \mathcal{F}_2\})$$

Note: The book's use of $\times$ is a slight abuse of notation. Rigorous texts use $\mathcal{F}_1 \otimes \mathcal{F}_2$ to distinguish the generated σ-field from a mere Cartesian product of sets.

Theorem 6.2 (**Generators and Projections**). *(i) The product σ-field $\mathcal{F}_1 \times \mathcal{F}_2$ is generated by the 'cylinders': $\mathcal{C} = \{A_1 \times \Omega_2\} \cup \{\Omega_1 \times A_2\}$.*

(ii) $\mathcal{F}_1 \times \mathcal{F}_2$ is the smallest σ-field making the projections $Pr_i : \Omega \to \Omega_i$ measurable.

Crux. (i) Cylinders are rectangles, so $\mathcal{C} \subset \mathcal{R} \implies \sigma(\mathcal{C}) \subseteq \sigma(\mathcal{R})$ by Corollary 2.26. Conversely, $A_1 \times A_2 = (A_1 \times \Omega_2) \cap (\Omega_1 \times A_2) \implies \mathcal{R} \subset \sigma(\mathcal{C}) \implies \sigma(\mathcal{R}) \subseteq \sigma(\mathcal{C})$. Thus $\sigma(\mathcal{R}) = \sigma(\mathcal{C})$.

(ii) Preimages of projections are exactly the cylinders ($\mathrm{Pr}_1^{-1}(A_1) = A_1 \times \Omega_2$). Thus, making projections measurable strictly requires generating $\sigma(\mathcal{C})$, which by (i) is $\mathcal{F}_1 \times \mathcal{F}_2$.

$\square$

Proposition 6.3. *On $\mathbb{R}^2$, the Borel σ-field generated by Borel rectangles $\mathcal{R} = \{B_1 \times B_2\}$ equals the σ-field generated by 2D intervals $\mathcal{I} = \{I_1 \times I_2\}$.*

Crux. Trivially, $\mathcal{I} \subset \mathcal{R} \implies \sigma(\mathcal{I}) \subseteq \sigma(\mathcal{R})$ by Corollary 2.26.

Conversely, define the "good sets" $\mathcal{G} = \{A \subseteq \mathbb{R} : A \times \mathbb{R} \in \sigma(\mathcal{I})\}$. $\mathcal{G}$ is a σ-field containing all 1D intervals, implying $\mathcal{B} \subseteq \mathcal{G}$. Thus, for any $B_1, B_2 \in \mathcal{B}$, the cylinders $B_1 \times \mathbb{R}$ and $\mathbb{R} \times B_2$ are in $\sigma(\mathcal{I})$. Their intersection is $B_1 \times B_2 \in \sigma(\mathcal{I}) \implies \mathcal{R} \subset \sigma(\mathcal{I}) \implies \sigma(\mathcal{R}) \subseteq \sigma(\mathcal{I})$ again by Corollary 2.26. □

6.3 Construction of the product measure

Definitions: Sections & σ-finite

- **σ-finite:** A measure P on $(\Omega, \mathcal{F})$ is σ-finite if $\Omega = \bigcup_{n=1}^{\infty} A_n$ with $P(A_n) < \infty$.
- **Section:** For $A \subseteq \Omega_1 \times \Omega_2$, the ω_2-section is $A_{\omega_2} \triangleq \{\omega_1 \in \Omega_1 : (\omega_1, \omega_2) \in A\}$.
- **Slicing Commutativity:** Sections perfectly commute with all set operations:

$$(\Omega \setminus A)_{\omega_2} = \Omega_1 \setminus A_{\omega_2}, \quad \left(\bigcup_n A_n\right)_{\omega_2} = \bigcup_n (A_n)_{\omega_2}, \quad \left(\bigcap_n A_n\right)_{\omega_2} = \bigcap_n (A_n)_{\omega_2}.$$

- **Global-Local Equivalence:** Global relations hold if and only if they hold strictly across all possible slices. For example: $A \subseteq B \iff A_{\omega_2} \subseteq B_{\omega_2}\ \forall \omega_2 \in \Omega_2$, and $A \cap B = \emptyset \iff A_{\omega_2} \cap B_{\omega_2} = \emptyset\ \forall \omega_2 \in \Omega_2$.

Theorem 6.4 (**Measurability of Sections**)**.** *If $A \in \mathcal{F}_1 \times \mathcal{F}_2$, then every section $A_{\omega_2} \in \mathcal{F}_1$ and $A_{\omega_1} \in \mathcal{F}_2$.*

Crux. Let $\mathcal{G} = \{A \in \mathcal{F} : A_{\omega_2} \in \mathcal{F}_1\}$. (i) **Generators:** For $A_1 \times A_2$, A_{ω_2} is A_1 (if $\omega_2 \in A_2$) or $\emptyset$. Both $\in \mathcal{F}_1$. (ii) **σ-field:** Slicing commutes with complements and countable unions. Thus $\mathcal{G}$ is a σ-field containing all rectangles $\implies \mathcal{G} = \mathcal{F}_1 \times \mathcal{F}_2$. □

Definitions: Fields & Monotone Classes

- **Field ($\mathcal{A}$):** Closed under *finite* unions and complementation.
- **Monotone Class ($\mathcal{G}$):** A family of sets closed under countable increasing unions ($A_n \uparrow \implies \cup A_n \in \mathcal{G}$) and countable decreasing intersections ($A_n \downarrow \implies \cap A_n \in \mathcal{G}$).
- **MCT Application:** To prove a property holds for all sets in a σ-field $\mathcal{F}$ generated by a field $\mathcal{A}$ whose elements satisfy the property already, define $\mathcal{G} = \{A \in \mathcal{F} : A \text{ satisfies property}\}$. Show $\mathcal{G}$ is a monotone class containing the generating field $\mathcal{A}$, **which by MCT implies $\mathcal{G} = \mathcal{F}$.**

Theorem 6.7 (**Monotone Class Theorem**)**.** *The smallest monotone class $\mathcal{G}_{\mathcal{A}}$ containing a field $\mathcal{A}$ coincides with $\sigma(\mathcal{A})$.*

Crux. Trivially $\mathcal{G}_{\mathcal{A}} \subseteq \sigma(\mathcal{A})$. To prove $\mathcal{G}_{\mathcal{A}}$ is a σ-field:

(i) **Closure under Complementation:** $\mathcal{C} = \{A : A^c \in \mathcal{G}_{\mathcal{A}}\}$ is a monotone class containing $\mathcal{A} \implies \mathcal{G}_{\mathcal{A}} \subseteq \mathcal{C}$.

(ii) **Field Ext 1:** Fix $A \in \mathcal{A}$. $\mathcal{D} = \{B : A \cup B \in \mathcal{G}_{\mathcal{A}}\}$ is a monotone class containing $\mathcal{A} \implies \mathcal{G}_{\mathcal{A}} \subseteq \mathcal{D}$.

(iii) **Field Ext 2:** Fix $G \in \mathcal{G}_{\mathcal{A}}$. $\mathcal{E} = \{A : G \cup A \in \mathcal{G}_{\mathcal{A}}\}$ is a monotone class. By (ii), $\mathcal{A} \subseteq \mathcal{E} \implies \mathcal{G}_{\mathcal{A}} \subseteq \mathcal{E}$. Thus $\mathcal{G}_{\mathcal{A}}$ is a field.

(iv) **Promotion:** A field closed under increasing limits is a σ-field. Any countable union $\bigcup A_n$ can be written as the increasing limit of finite unions $B_n = \bigcup_{i=1}^n A_i$. Thus $\mathcal{G}_{\mathcal{A}}$ is a σ-field, so $\sigma(\mathcal{A}) \subseteq \mathcal{G}_{\mathcal{A}}$.

$\square$

Theorem 6.5. *Suppose that P_1, P_2 are finite measures. If $A \in \mathcal{F}$, then the functions*

$$\omega_2 \to P_1(A_{\omega_2}), \quad \omega_1 \to P_2(A_{\omega_1})$$

are measurable with respect to $\mathcal{F}_1$ and $\mathcal{F}_2$, respectively, and

$$\int_{\Omega_2} P_1(A_{\omega_2})\,\mathrm{d}P_2(\omega_2) = \int_{\Omega_1} P_2(A_{\omega_1})\,\mathrm{d}P_1(\omega_1).$$

ERRATA: The text has swapped the sets of integration Ω_1 and Ω_2!!

Crux. Let $\mathcal{G} = \{A \in \mathcal{F} : \text{theorem holds for } A\}$.

(i) **Base:** For $A_1 \times A_2$, $P_1(A_{\omega_2}) = P_1(A_1)\mathbb{I}_{A_2}(\omega_2)$. Integral evaluates to $P_1(A_1)P_2(A_2)$ symmetrically.

(ii) **Field ($\mathcal{A}$):** $\mathcal{A}$ strictly consists of finite disjoint unions of rectangles (a fact that we take for granted without proof) so any $B \in \mathcal{A}$ can be expressed as $B = \bigcup_{i=1}^n (A_{1i} \times A_{2i})$ where the basic rectangles are strictly disjoint. Slicing commutes with disjoint unions so we have

$$P_1(B_{\omega_2}) = \sum_{i=1}^n P_1((A_{1i} \times A_{2i})_{\omega_2}) = \sum_{i=1}^n P_1(A_{1i})\mathbb{I}_{A_{2i}}(\omega_2).$$

This is a finite sum of indicators, hence a measurable simple function. By the linearity of the integral, integrating $P_1(B_{\omega_2})$ over Ω_2 yields $\sum_{i=1}^n P_1(A_{1i})P_2(A_{2i})$. The reverse integral yields the identical sum so that $\mathcal{A} \subset \mathcal{G}$.

(iii) **MC (↑):** Let $\{B_k\} \uparrow B$. The sections $(B_k)_{\omega_2} \uparrow B_{\omega_2}$. By continuity of measure from below, $P_1((B_k)_{\omega_2}) \uparrow P_1(B_{\omega_2})$. As a pointwise limit of measurable functions, $P_1(B_{\omega_2})$ is measurable. By the Monotone Convergence Theorem (MCT):

$$\int_{\Omega_2} P_1(B_{\omega_2})\,\mathrm{d}P_2(\omega_2) = \lim_{k\to\infty} \int_{\Omega_2} P_1((B_k)_{\omega_2})\,\mathrm{d}P_2(\omega_2).$$

Applying MCT to the reverse integral yields the same limit. Thus, $B \in \mathcal{G}$.

(iv) **MC (↓):** Let $\{B_k\} \downarrow B$. Since P_1 is finite, measure continuity from above holds $P_1((B_k)_{\omega_2}) \downarrow P_1(B_{\omega_2})$. Because P_2 is finite, the integrals are bounded by $P_1(\Omega_1)P_2(\Omega_2) < \infty$. Applying (DCT), we conclude that the limits of the integrals converge and are equal. Thus, $B \in \mathcal{G}$.

By Theorem 6.7, $\sigma(\mathcal{A}) = \mathcal{F} \subseteq \mathcal{G}$. □

Definition 6.Z (Product Measure): For finite measure spaces,

$$P(A) \triangleq \int_{\Omega_2} P_1(A_{\omega_2})\,\mathrm{d}P_2(\omega_2).$$

Theorem 6.8. *The product measure P is countably additive and uniquely determined by its values on rectangles.*

Crux. **Additivity:** For disjoint A_n, slicing commutes with unions. Apply MCT to swap integral and sum:

$$\begin{aligned} P\left(\bigcup_{n=1}^{\infty} A_n\right) &= \int_{\Omega_2} P_1\left(\bigcup_{n=1}^{\infty} (A_n)_{\omega_2}\right)\mathrm{d}P_2(\omega_2) \\ &= \int_{\Omega_2} \sum_{n=1}^{\infty} P_1((A_n)_{\omega_2})\,\mathrm{d}P_2(\omega_2) = \sum_{n=1}^{\infty} P(A_n). \end{aligned}$$

Uniqueness: Let Q be another product measure. Let $\mathcal{G} = \{A \in \mathcal{F} : P(A) = Q(A)\}$. $\mathcal{G}$ contains the field of finite disjoint rectangles. Because P and Q are finite measures, continuity of measure from below and above guarantees that if $A_n \uparrow A$ or $A_n \downarrow A$ in $\mathcal{S}$, the limit A is also in $\mathcal{S}$. Thus $\mathcal{G}$ is a monotone class $\implies \mathcal{G} = \mathcal{F}$.

□

Note: The "Completion Trap" for Product Spaces

Let $\mathcal{B}$ be the 1D Borel σ-field and $\mathcal{M}$ be the 1D Lebesgue σ-field (the completion of $\mathcal{B}$). While taking the product of Borel sets works cleanly ($\mathcal{B} \times \mathcal{B} = \mathcal{B}(\mathbb{R}^2)$), taking the product of Lebesgue sets creates a structurally incomplete space: $\mathcal{M} \times \mathcal{M} \neq \mathcal{M}_2$.

- **The Pathology:** Let $A \notin \mathcal{M}$ be a non-Lebesgue-measurable set (e.g., a Vitali set) on the y-axis. Let $N = \{0\}$ be a single point on the x-axis, which is a null set ($N \in \mathcal{M}, m(N) = 0$).
- **The Contradiction:** Consider the 2D vertical line $N \times \mathbb{R} \in \mathcal{M} \times \mathcal{M}$. Its product measure is strictly $0 \times \infty = 0$. In a complete 2D space, any subset of this null line, such as $E = \{0\} \times A$, *should* be measurable and have measure zero. However, if $E \in \mathcal{M} \times \mathcal{M}$, then by Fubini's section theorem, its 1D section at $x = 0$ must be measurable in $\mathcal{M}$. Since that section is exactly $A \notin \mathcal{M}$, the set E is strictly invisible to $\mathcal{M} \times \mathcal{M}$.
- **The Fix:** To obtain the true 2D Lebesgue σ-field $\mathcal{M}_2$, we must manually "complete" $\mathcal{M} \times \mathcal{M}$ by taking every subset of every 2D measure-zero set and explicitly appending it to the σ-field.

6.4 Fubini's theorem

Theorem 6.10. *If a function $f : \Omega_1 \times \Omega_2 \to \mathbb{R}$ is measurable with respect to $\mathcal{F}_1 \times \mathcal{F}_2$, then for each $\omega_1 \in \Omega_1$, the section function $f_{\omega_1}(\omega_2) \triangleq f(\omega_1, \omega_2)$ is $\mathcal{F}_2$-measurable. (The symmetric result holds for ω_2).*

Crux. For any Borel set $B \in \mathcal{B}(\mathbb{R})$, taking a section and an inverse image commute:

$$(f_{\omega_1})^{-1}(B) = \{\omega_2 : f(\omega_1, \omega_2) \in B\} = (f^{-1}(B))_{\omega_1}$$

Because f is measurable, $f^{-1}(B) \in \mathcal{F}_1 \times \mathcal{F}_2$. By Theorem 6.4, the ω_1-section of any 2D measurable set is in $\mathcal{F}_2$. Thus, $(f_{\omega_1})^{-1}(B) \in \mathcal{F}_2$. □

Corollary 6.10. *For any measurable function f, sectioning and taking inverse images are commutative: $(f_{\omega_1})^{-1}(B) = (f^{-1}(B))_{\omega_1}$.*

Corollary 6.11. *For a measurable, non-negative function $f : \Omega_1 \times \Omega_2 \to \mathbb{R}$, the marginal integral functions*

$$G(\omega_1) \triangleq \int_{\Omega_2} f(\omega_1, \omega_2)\, \mathrm{d}P_2(\omega_2), \qquad H(\omega_2) \triangleq \int_{\Omega_1} f(\omega_1, \omega_2)\, \mathrm{d}P_1(\omega_1)$$

are $\mathcal{F}_1$ and $\mathcal{F}_2$-measurable, respectively.

Crux. Apply the Standard Constructive Machine:

(i) **Indicators:** $f = \mathbb{I}_A \implies G(\omega_1) = P_2(A_{\omega_1})$, which is $\mathcal{F}_1$-measurable by Theorem 6.5.

(ii) **Simple:** $f = \sum c_i \mathbb{I}_{A_i} \implies G(\omega_1)$ is measurable by linearity.

(iii) **Non-Negative:** $f_n \uparrow f \implies G_n(\omega_1) \uparrow G(\omega_1)$ by MCT. Pointwise limits of measurable functions are measurable.

$\square$

Notes on Integration:

- **General f:** Decompose $f = f^+ - f^-$. $G(\omega_1)$ is measurable provided the marginal integrals of f^+ and f^- are strictly finite, completely avoiding the $\infty - \infty$ trap.
- **Product Measure:** $(\Omega_1 \times \Omega_2, \mathcal{F}_1 \times \mathcal{F}_2, P_1 \times P_2)$ is a standard measure space. The 2D integral $\int f \, \mathrm{d}(P_1 \times P_2)$ strictly uses the standard 1D constructive machinery.

Theorem 6.12 (**Tonelli's Theorem**). *Let f be a measurable non-negative function defined on $\Omega_1 \times \Omega_2$. Then*

$$\begin{aligned}\int_{\Omega_1 \times \Omega_2} f(\omega_1, \omega_2) \, \mathrm{d}(P_1 \times P_2)(\omega_1, \omega_2) &= \int_{\Omega_1} \left(\int_{\Omega_2} f(\omega_1, \omega_2) \, \mathrm{d}P_2(\omega_2) \right) \mathrm{d}P_1(\omega_1) \\ &= \int_{\Omega_2} \left(\int_{\Omega_1} f(\omega_1, \omega_2) \, \mathrm{d}P_1(\omega_1) \right) \mathrm{d}P_2(\omega_2). \qquad (6.4)\end{aligned}$$

Crux. Apply the Standard Constructive Machine:

(i) **Indicators:** $f = \mathbb{I}_E \implies$ Equality is exactly the definition of Product Measure (Def 6.Z).

(ii) **Simple:** Equality holds by integral linearity.

(iii) **Non-Negative:** $f_n \uparrow f$. Apply MCT to both the 2D integral and the nested 1D integrals to pass the limit through all integral signs.

$\square$

Theorem 6.13 (**Fubini's Theorem**). *If $f \in L^1(\Omega_1 \times \Omega_2, P_1 \times P_2)$, then for P_1-almost all ω_1 and P_2-almost all ω_2, the sections f_{ω_1} and f_{ω_2} are integrable. Furthermore, the marginal integrals*

$$G(\omega_1) \triangleq \int_{\Omega_2} f(\omega_1, \omega_2) \, \mathrm{d}P_2(\omega_2), \qquad H(\omega_2) \triangleq \int_{\Omega_1} f(\omega_1, \omega_2) \, \mathrm{d}P_1(\omega_1)$$

are in $L^1(\Omega_1, P_1)$ and $L^1(\Omega_2, P_2)$, respectively, and 6.4 holds: in concise form it reads

$$\int_{\Omega_1 \times \Omega_2} f \, \mathrm{d}(P_1 \times P_2) = \int_{\Omega_1} \left(\int_{\Omega_2} f \, \mathrm{d}P_2 \right) \mathrm{d}P_1 = \int_{\Omega_2} \left(\int_{\Omega_1} f \, \mathrm{d}P_1 \right) \mathrm{d}P_2.$$

Crux. Decompose $f = f^+ - f^-$. Since $f \in L^1$, both components are strictly non-negative and integrable ($< \infty$). Apply Tonelli's Theorem to f^+ and f^- individually. Subtracting their Tonelli equations legally avoids the $\infty - \infty$ trap as $f \in L^1$. Linearity yields the result. □

Proposition 6.16. *Let $f : \mathbb{R} \to \mathbb{R}$ be measurable and positive. Consider the set of all points in the upper half-plane being below the graph of f:*

$$A_f \triangleq \{(x, y) : 0 \leq y \leq f(x)\}.$$

Then A_f is m_2-measurable and $m_2(A_f) = \int f(x)\,\mathrm{d}x$.

Crux. **Measurability:** $F(x, y) = f(x) - y$ is m_2-measurable. $A_f = \{y \geq 0\} \cap \{F(x, y) \geq 0\}$ is the intersection of measurable inverse images, hence m_2-measurable. **Measure:** Apply Tonelli to the indicator $\mathbb{I}_{A_f}$:

$$m_2(A_f) = \int_{\mathbb{R}} \left(\int_{\mathbb{R}} \mathbb{I}_{A_f}(x, y)\,\mathrm{d}y \right) \mathrm{d}x = \int_{\mathbb{R}} m_1([0, f(x)])\,\mathrm{d}x = \int_{\mathbb{R}} f(x)\,\mathrm{d}x.$$

□

6.5 Probability

Definitions: Random Vectors, Distributions & Densities

- **Measurability:** A random vector $Z = (X, Y) : \Omega \to \mathbb{R}^2$ is $\mathcal{F}$-measurable if the pre-image of any 2D Borel set is in $\mathcal{F}$.
- **Joint Distribution:** $P_{(X,Y)}(B) \triangleq P(Z \in B)$ for $B \in \mathcal{B}(\mathbb{R}^2)$.
- **Joint Density:** A random vector (X, Y) is said to have a joint density if there exists a non-negative function $f_{(X,Y)} \in L^1(\mathbb{R}^2)$ such that $P_{(X,Y)}(B) = \int_B f_{(X,Y)}\,\mathrm{d}m_2$ for every 2D Borel set $B \in \mathcal{B}(\mathbb{R}^2)$.
- **Marginals:** $P_X(A) = P_{(X,Y)}(A \times \mathbb{R})$. By Tonelli, marginal densities are found by integrating out the alternate variable: $f_X(x) = \int_{\mathbb{R}} f_{(X,Y)}(x, y)\,\mathrm{d}y$.
- **Gaussian Density:** $n(x, y) = \frac{1}{2\pi\sqrt{1-\rho^2}} \exp\left\{ -\frac{x^2 - 2\rho xy + y^2}{2(1-\rho^2)} \right\}$.

Proposition 6.16.5. If $X, Y : \Omega \to \mathbb{R}$ are $\mathcal{F}$-measurable, then the vector $Z = (X, Y) : \Omega \to \mathbb{R}^2$ is $\mathcal{F}/\mathcal{B}(\mathbb{R}^2)$-measurable.

Notation Note ($\mathcal{F}_1/\mathcal{F}_2$-measurability): Formally, a function $f : (\Omega_1, \mathcal{F}_1) \to (\Omega_2, \mathcal{F}_2)$ is called $\mathcal{F}_1/\mathcal{F}_2$-*measurable*. By convention, if the target space is $\mathbb{R}$ or $\mathbb{R}^d$ equipped with its standard Borel σ-field $\mathcal{B}$, the target σ-field is omitted and f is simply called $\mathcal{F}_1$-*measurable*.

Crux. The Borel σ-field $\mathcal{B}(\mathbb{R}^2)$ is generated by measurable rectangles $A \times B$. The pre-image of a rectangle is:

$$Z^{-1}(A \times B) = \{\omega \in \Omega : X(\omega) \in A, Y(\omega) \in B\} = X^{-1}(A) \cap Y^{-1}(B).$$

Since X and Y are $\mathcal{F}$-measurable, this is a finite intersection of sets in $\mathcal{F}$, which strictly guarantees $Z^{-1}(A \times B) \in \mathcal{F}$. $\square$

Theorem 6.19 (**Convolution Integral**)**.** *If X, Y have joint density $f_{(X,Y)}$, the density of $X + Y$ is:*

$$f_{X+Y}(z) = \int_{\mathbb{R}} f_{(X,Y)}(x, z - x)\,\mathrm{d}x. \tag{6.6}$$

Crux. $F_{X+Y}(z) = P(X + Y \leq z) = \int_{\mathbb{R}} \left(\int_{-\infty}^{z-x} f_{(X,Y)}(x, y)\,\mathrm{d}y \right) \mathrm{d}x$ via Tonelli. Substitute $v = x + y \implies \mathrm{d}v = \mathrm{d}y$, pushing the z bound to the outer integral. Differentiating w.r.t z yields 6.6. $\square$

Theorem 6.20. *X, Y are independent $\iff P_{(X,Y)} = P_X \times P_Y$ on $\mathcal{B}(\mathbb{R}^2)$.*

Crux. ($\Rightarrow$) Independence implies $P_{(X,Y)}(B_1 \times B_2) = P_X(B_1)P_Y(B_2) = (P_X \times P_Y)(B_1 \times B_2)$. Let $\mathcal{G} = \{B \in \mathcal{B}(\mathbb{R}^2) : P_{(X,Y)}(B) = (P_X \times P_Y)(B)\}$. $\mathcal{G}$ contains the field $\mathcal{A}$ of finite disjoint rectangles. Because probability measures are continuous from above/below, $\mathcal{G}$ is a monotone class $\implies \mathcal{G} = \mathcal{B}(\mathbb{R}^2)$.
($\Leftarrow$) Restrict B strictly to rectangles to recover independence definition. $\square$

Theorem 6.21. *If X, Y have a joint density, then they are independent if and only if*

$$f_{(X,Y)}(x, y) = f_X(x) f_Y(y) \quad m_2\text{-a.e.} \tag{6.7}$$

If X and Y are absolutely continuous and independent, then they have a joint density and it is given m_2-a.e. by 6.7.

Crux. ($\Rightarrow$) By independence and Tonelli, $\int_{A \times B} f_{(X,Y)}\,\mathrm{d}m = P(X \in A)P(Y \in B) = \int_{A \times B} f_X f_Y\,\mathrm{d}m$. By the Monotone Class Theorem, these integrals agree on all Borel sets $\implies$ integrands agree m_2-a.e.
($\Leftarrow$) Reverse Tonelli integration over rectangles recovers $P_X(A)P_Y(B)$.
Finally, if X and Y are independent and absolutely continuous, defining $g(x, y) = f_X(x) f_Y(y)$ perfectly satisfies the joint density criteria via MCT. $\square$

Corollary 6.22 & Proposition 6.23.

- If Gaussian variables are uncorrelated ($\rho = 0$), the exponential factors perfectly $\implies f_{(X,Y)} = f_X f_Y \implies$ independent.

- **Independent Convolution:** Substitute $f_{(X,Y)} = f_X f_Y$ into 6.6 to get $f_{X+Y}(z) = \int_{\mathbb{R}} f_X(x) f_Y(z-x)\,\mathrm{d}x$.

Theorem 6.24 (Inversion Formula). *If F_X is continuous at $a, b \in \mathbb{R}$ $(a < b)$:*

$$F_X(b) - F_X(a) = \lim_{c\to\infty} \frac{1}{2\pi} \int_{-c}^{c} \frac{e^{-iua} - e^{-iub}}{iu} \varphi_X(u)\,\mathrm{d}u.$$

Crux. Let J_c be the integral inside the limit. Substituting $\varphi_X(u) = \int_{\mathbb{R}} e^{iux}\,\mathrm{d}P_X(x)$ creates a double integral. The integrand modulus is bounded by $(b-a)$, hence it is L^1 on $[-c, c] \times \mathbb{R}$. Fubini permits swapping the order of integration:

$$J_c = \int_{\mathbb{R}} \underbrace{\left(\frac{1}{2\pi} \int_{-c}^{c} \frac{e^{iu(x-a)} - e^{iu(x-b)}}{iu}\,\mathrm{d}u \right)}_{I(x,c)} \mathrm{d}P_X(x).$$

Here, $I(x, c)$ is strictly the inner integral with respect to u. Using Euler's formula, the imaginary cosine terms are odd functions over the symmetric interval $[-c, c]$ and evaluate to zero. The real sine terms split via substitution $(y = u(x-a), z = u(x-b))$ to yield:

$$I(x,c) = \frac{1}{2\pi} \int_{-c(x-a)}^{c(x-a)} \frac{\sin y}{y}\,\mathrm{d}y - \frac{1}{2\pi} \int_{-c(x-b)}^{c(x-b)} \frac{\sin z}{z}\,\mathrm{d}z.$$

The Dirichlet integral $\int_{-T}^{T} \frac{\sin t}{t}\,\mathrm{d}t$ is globally bounded $\implies$ $|I(x, c)| \le M$. Since $\int_{\mathbb{R}} M\,\mathrm{d}P_X < \infty$, DCT applies. The pointwise limit $f(x) = \lim I(x, c)$ is the simple function $\mathbb{I}_{(a,b)}(x) + \frac{1}{2}\mathbb{I}_{\{a,b\}}(x)$. Since F_X is continuous at a, b, singletons are null sets $\implies$ $f(x) = \mathbb{I}_{(a,b)}(x)$ P_X-a.e.

Integrating this pointwise limit yields the final result:

$$\lim_{c\to\infty} J_c = \int_{\mathbb{R}} \mathbb{I}_{(a,b)}(x)\,\mathrm{d}P_X(x) = P_X((a, b)) = F_X(b) - F_X(a).$$

□

Corollary 6.25. *The characteristic function determines the probability distribution.*

Crux. Let P_1, P_2 be probability measures sharing φ_X. Their CDFs F_1, F_2 have at most countable discontinuities. For any $(a, b]$, choose a point of continuity $d \in (a, b)$ and sequences of continuity points $y_n \downarrow b$, $z_n \downarrow a$. The Inversion Formula equates $F_1(y_n) - F_1(d) = F_2(y_n) - F_2(d)$. Taking limits $n \to \infty$ leverages right-continuity, proving $P_1((d, b]) = P_2((d, b])$. We can similarly prove that $P_1((a, d]) = P_2((a, d])$ so that $P_1((a, b]) = P_2((a, b])$.

Let $\mathcal{G} = \{B \in \mathcal{B}(\mathbb{R}) : P_1(B) = P_2(B)\}$. $\mathcal{G}$ contains the field $\mathcal{A}$ of finite disjoint intervals. Continuity of finite measures from above/below guarantees $\mathcal{G}$ is a monotone class $\implies$ $\mathcal{G} = \mathcal{B}(\mathbb{R})$. □

Theorem 6.26. *If $\varphi_X \in L^1(\mathbb{R})$, then X has an absolutely continuous, bounded density:*

$$f_X(x) = \frac{1}{2\pi} \int_{-\infty}^{\infty} e^{-iux} \varphi_X(u) \, \mathrm{d}u.$$

Crux. Since the integrand modulus $\left|e^{-iux}\varphi_X(u)\right|$ is dominated by the L^1 function $|\varphi_X(u)|$, DCT yields

$$\lim_{n\to\infty} f_X(x_n) = \frac{1}{2\pi} \int_{-\infty}^{\infty} \lim_{n\to\infty} e^{-iux_n} \varphi_X(u) \, \mathrm{d}u = f_X(x).$$

Thus, f_X is continuous (and uniformly bounded by $\frac{1}{2\pi} \|\varphi_X\|_1$), and hence integrable on finite intervals. Because $\varphi_X \in L^1(\mathbb{R})$, Fubini allows integral swapping on finite intervals $[a, b] \times \mathbb{R}$. The Inversion Formula evaluates this to $\int_a^b f_X(x) \, \mathrm{d}x = F_X(b) - F_X(a)$ for continuity points.

By Lemma 4.16.66 (with $I = \mathbb{R}$), boundedness of $f_X \implies$ its integral is continuous everywhere w.r.t a, b. Since right-continuous F_X matches a continuous function on a dense set, F_X is forced continuous everywhere. Monotonicity implies $f_X \geq 0$ a.e. Limits $a \to -\infty, b \to \infty$ yield $\int_{-\infty}^{\infty} f_X(x) \, \mathrm{d}x = 1$. $\square$